

Reinforcement Learning for Stock Market Trading: A Framework for Predictive and Adaptive Decision-Making

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Abstract

The extreme dynamism and non-stationarity of financial markets pose a basic difficulty to classical quantitative predictive models, which are not always very adaptive to changing economic regimes. In this case study, we deal with this volatility by examining how to apply Reinforcement Learning (RL), which is a naturally well-posed solution when faced with sequential decision-making in uncertainty, a trading modeling as a Markov Decision Process (MDP). In contrast to supervised, RL agents improve the best policies by implementing long-term cumulative reward signals by interacting with the market environment in a continuous manner. The preliminary study proved that Deep Reinforcement Learning (DRL) agents, including Proximal Policy Optimisation (PPO) and Deep Q-Networks (DQN) can learn profitable behaviors that can outperform their static baseline counterparts. The essence of this work is the creation of an elaborate DRL framework that aims at robustness and financial congruence. The framework proposed combines three methodological innovations: continuous action space and hybrid action space are applied to multi-asset portfolio allocation, dynamic DRL ensemble strategies are implemented to address the issue of market non-stationarity, and composite and risk-adjusted reward functions are used to explicitly seek to maximize capital preservation and high Sharpe ratios.

Keywords

Reinforcement Learning, Deep learning, Algorithmic Trading, Financial Markets, Portfolio collecting, Ensemble methods, Continuous Action Space, Risk adjusted reward, Proximal Policy Optimisation (PPO).

I. Introduction

A. History of Reinforcement Learning in Financial Markets.

There has been a rapid increase in the use of Artificial Intelligence (AI) and Machine Learning (ML) in financial trading, and Deep Reinforcement Learning (DRL) has become a promising paradigm in the development of autonomous trading systems. According to RW, the algorithmic trading problem can be seen as an interaction between an

An Environment (the stock market), and an Agent (the trading algorithm), formalised as a Markov Decision Process (MDP). The agent in this model monitors the state (S), acts (A) and is rewarded (R), is taught an optimal policy (π) to maximize the expected cumulative reward over time.

Directly learning policies with DRL with high-dimensional input features, including raw price data, technical indicators and portfolio status gives it a significant edge over deterministic predictor models of the rules. Value-based DRL algorithms such as DQN, policy gradient algorithms such as PPO, Advantage Actor-Critic (A2C), and Deep Deterministic Policy Gradient (DDPG) have been effectively applied to a wide range of financial problems, such as optimal order execution, portfolio management, and algorithmic trading. It has been demonstrated empirically that such DRL algorithms can actually outperform classical time series momentum strategies and traditional portfolio allocation methods by changing with the market conditions.

B. Significance and Relevancy of the Problem.

The stock market is highly complex, volatile and non-stationary, i.e. the statistical properties of the data vary with time. Classical quantitative frameworks based on assumptions about data distributions tend to fail as market regimes change (e.g. between bull and bear markets). This lack of adaptation causes the degradation of performance and obsolescence of the models.

Relevance of DRA is that it has the ability to adaptively learn and optimize in a sequence. With constant adaptation of its strategy according to cumulative performance and risk indicators, an RL agent is able to operate in complex market microstructures and dynamically invest capital. The efficient application of RL will deliver an automated and systematic platform of creating risk-adjusted sustainable returns. But the weaknesses of DRL policies, especially their susceptibility to hyperparameters and their overfitting of turbulent historical data, represent major risks, and are currently the main reason why they are not used more intensively in high-stakes financial tasks. Thus, there is a need to have a sound, production-ready structure to help in closing the gap between theory and practice.

C. Objective of the Study

The ultimate aim of this research is to go beyond basic DRL application validation, and to construct a solid, conceptual DRL platform of advanced, multi-asset algorithmic trading.

The specific aims are to:
Critically examine existing DRL systems in finance by architecture (value, policy) and point out the existing weaknesses (e.g., action space, overfitting).

Establish and rationalize a holistic framework that includes approaches to address the problems identified, emphasizing on the risk management and policy stability.

State: Specify the architecture of a proposed agent, including the most important components, which are State, Action, Reward, and Environment to make sound multi-asset portfolio optimization and fractional trading decisions.

II. Literature Review

Applications Deep Reinforcement Learning (DRL) to finance can be broadly categorized in terms of its algorithmic structure that points to the replacement

of simple discrete trading decisions by complex continuous control problems including portfolio allocation.

A. DRAMAT DRL Architecture Categorization.

- ***Value-Based Approaches:***

Value-Based methods were the initial ones to attain success, the most renowned of which is Deep Q-Networks (DQN), a neural network that approximates the optimal state-action value function $Q(s,a)$. This architecture can be applied to a high dimensional state space which financial data has but is inherently limited to discrete action space (e.g. fixed Buy, Sell or Hold decisions). This drawback precludes its application to the problems where capital must be allocated fine and fractionally, i.e. in multi-asset portfolio management, in, etc.

- ***Policy-Based and Actor-Critic Methods:***

Continuing on the limitations of discrete action space, research changed focus to policy gradient methods, most notably to the Actor-Critic family. Such methods find the optimal policy $\pi(a|s)$ that provides a mapping between states and actions. Proximal Policy Optimisation (PPO) has attracted a large following due to its stability but at the cost of clipping the updates of policy with an objective function denoted. In continuous control applications, like those required in optimal fractional trading and portfolio allocation, algorithms like

Deep Deterministic Policy Gradient (DDPG) and Soft ActorCritic (SAC) are moreover required as they are specifically designed to generate continuous action vectors (e.g., portfolio weights) , . Studies confirm that such continuous action algorithms are necessary to provide the best returns on a large portfolio of assets.

Robustness and Ensemble Strategies: A RL agent that would optimal perform in every regime of the market (bull, bear or volatile) cannot be guaranteed to behave optimally in all non-stationary financial markets and thus, to enhance robustness and reduce model brittle nature, current literature aims at. Ensemble DRL

strategies . These combinations combine the results of multiple specialized agents, typically PPO, A2C and DDPG . The idea is to train these agents simultaneously, and then dynamically select the best performing policy with a quantitative risk-adjusted measure, like the Sharpe Ratio, during a rolling validation time horizon. The backtests of the empirical Dow Jones stock data shows that such ensemble approach produces far superior risk-adjusted returns than any of the DRL algorithms or baseline models.

B. Problems and Literature Weaknesses.

Even though it has been proven during simulations, it has three gaps that are critical to the application of DRL agents to production finance:

Continuous-Discrete Duality Continuous action spaces (DDPG or SAC) are needed to enable generalization across assets as well as in fractional sizing, but discrete market tick precision, i.e. must allow orders to be finalized at discrete market tick sizes, . This adds some sort of continuity discrete duality that limits execution efficiency. Current solutions propose

Hybrid RL schemes where an agent with continuous time is optimizing the optimal level of trading, and a second agent, a discrete time agent, is optimizing the actual limit price, a combination of generalization and specialization, .

Reward Misalignment and Risk: Rewards policies trained to maximize either short-term or cumulative reward often promote reward hacking and risk-seeking, which generate excessive drawdowns, To balance the RL goal with a body of established financial utility theory, such as the Modern Portfolio Theory, composite reward functions are proposed to explicitly trade-off both return and risk. The penalties of these functions are risk-adjusted like the.

Sortino Ratio (only downside volatility) or the Treynor Ratio, or the Market-Adaptive Ratio (MAR) (that scales its risk measurement flexibly based on whether market is bull or bear market regime),

Overfitting and Feature Quality Paradox: The hyperparameter optimization and the design of the

state representation are extremely vulnerable to the performance of DRL models. Studies show a

Feature Quality Paradox: excess technical indicators or excess history will introduce noise that overwhelm the agent, and will cause it to overfit non-transferable patterns, On the other hand, the addition of non-traditional, filtered signals, e.g. sentiment analysis based on financial news, can introduce a qualitative aspect to the trading that will be more successful and reduce the quantitative trading/qualitative trading gap, .

III. Dataset Description

The Kaggle Nifty50 Stocks Dataset, which serves as an alternative high-liquidity asset universe for multi-asset trading algorithms, provides comprehensive historical data for the top Indian stock market index.

- **Asset Universe:** The dataset contains the price history and trading volumes for the fifty stocks in the NIFTY 50 Index. The NIFTY 50 is a benchmark index reflecting the weighted average of 50 of the largest Indian companies listed on the National Stock Exchange (NSE) of India.
- **Data Source:** The original data is sourced from NSE India.
- **Data Level/Frequency:** The data is typically provided at a day-level.
- **Key Features (Columns):** The files include essential financial metrics, such as:
 - **Date:** The date of the trading session.
 - **Open:** The opening price of the stock on that day.
 - **High:** The highest price reached during the trading session.
 - **Low:** The lowest price reached during the trading session.
 - **Close:** The closing price of the stock on that day.
 - **Volume:** The total number of shares traded during the session.

- **Adj Close:** The adjusted closing price, which accounts for corporate actions like dividends and stock splits (varies by specific dataset version).
- **Duration:** Various versions of the dataset exist; a common long-term version covers the period from January 1, 2000, to April 30, 2021. This long duration is vital for training DRL agents across a wide range of market regimes, as specified in the research methodology.
- **Relevance to RL Framework:** Its use of highly liquid assets (the NIFTY 50 components) helps ensure that the simulated volume and execution costs are realistic, preventing the DRL agent from learning non-viable strategies based on unrealistic execution in illiquid markets.

IV. Problem Statement

A. Sufficient Delimiting of Research Gap.

The fundamental research gap that this framework attempts to fill is the absence of any single, unified and adaptive DRL framework that can sustain optimal risk-adjusted returns in the non-stationary, multi-asset trading environment. Single DRA agents are brittle to models, hyperparameter sensitive, and unable to dynamically change policy in response to changing market conditions (e.g. between low and high volatility regimes). Importantly, the current production strategies tend to either employ over simplified discrete action or not incorporate critical risk management (such as considering transaction costs) in the learning goal.

B. Statement of Problem Scope.

This research problem is the multi-asset portfolio allocation problem in high-liquidity equity markets (e.g. Dow Jones or S&P 500 components).

Problem: The agent has to decide on the best fractional weight distribution of a set of N assets at each period, to maximize the terminal portfolio value, and at the same time reduce systemic risk and maximum drawdown.

Constraint: The solution should be able to deal with continuous action spaces (fractional trading decisions) and be policy robust when faced with sudden shifts in market direction and volatility.

C. Reasons why Reinforcement Learning is Apposite.

To a certain degree, RL alone is particularly adapted to the trading problem owing to the sequential decision-making nature of the trading problem alongside the partial observability of the problem. Contrary to supervised learning, which merely predicts the next price level, RL addresses the cumulative effect of a course of action (Buy, Hold, Sell, Reallocate) in the long-term. This procedure also presupposes striking a balance:

Exploration vs. Exploitation: The agent has to explore new, which could be profitable strategies and exploit existing knowledge.

Dynamic Adaptation: The policy should be able to change with changes in the state without any assumptions of data stationarity and it is better than the traditional optimization strategies.

Utility Maximization: The reward hypothesis enables straightforward optimization of intricate monetary utility functions (e.g., the Market-Adaptive Ratio), which provides the goal of the AI with human financial targets (risk-adjusted return).

V. Methodology

The Simulation of the Environment and Data Preparation.

The experimental setup will be based on a multi-asset trading market (e.g. high liquidity stocks such as Dow Jones components) being simulated with historical price data (Open, High, Low, Close, Volume). The environment is designed based on an interface that resembles a gym, which makes it easy to interface with the standard DRL libraries. Another important part of the simulation consists of careful incorporation of realistic market microstructure effects, such as

into the reward and state dynamics of the environment directly, transaction costs and slippage. It is important to train the agent using such penalties to avoid overfitting to idealized simulations resulting in non-viable high-frequency trading policies.

B. Definition of State and Action Space.

1) State Representation (S)

The state S_t at time t a vector of features is provided to the agent that is based on three main categories of features:

Market Data: Normalized historical price data and technical indicators (TI) like Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD) and trading volume.

Portfolio Status: Internal data on the current financial status of the agent such as the cash available, current fractional holdings of stock in the agent and the time frame of the current trades.

Rationale of Feature Engineering: To avoid the so-called Feature Quality Paradox (high signal indicators combined with large numbers of correlated features leads to noise and overfitting) feature engineering will favor high-signal features and eliminate redundant features.

2) Action Space (A)

In the system, there is the use of a continuous action space in which the agent produces a portfolio weight vector $A_t = [w_1, w_2, \dots, w_N]$. These weights are the fractional distribution of capital in the N assets, and it needs $\sum w_i = 1$ (self-financing condition). Such a configuration requires Actor-Critic algorithms. To be executed perfectly, the framework will serve in support of a.

Hybrid RL scheme, in which a continuous agent decides the amount of traded, and a fine-grained agent chooses the ultimate, market-relevant limit price.

C. Design of Composite Reward Function.

The reward functional R is formulated as a mixed purpose so as to guarantee the agent acquires a technically sustainable, risk controlled policy, and compromise of return objectives and risk objectives. The role has 3 differentiable elements:

Component of return, Immediate portfolio multiple or differential return.

Risk Penalty: Implicit risk volatility penalties, such as an inverse of the Sortino Ratio (involving only downside risk), or as an inverse of σ .

Market-Adaptive Ratio (MAR) which adds a parameter ρ to characterise bull and bear market regimes.

Transaction Cost Penalty: A penalty based on the simulated cost of slippage and commission of carrying out the trade.

D. Training and Evaluating Process.

1) Ensemble Training

The system employs an Ensemble DRL approach which entails specialized actor/critic agents of PPO, A2C and DDPG/SAC. Training of conclusions of historical data on a growing time window is conducted in parallel on the agents.

2) Dynamic Policy Selection

The essence of the strength of the ensemble is commissioning an energetic policy. The entire collection of agents who have been trained after some period (say quarterly) is then tested on a hidden test window. The agent that shows greatest

Sharpe Ratio at this period of validation is dynamically chosen in order to make the trades of the next real or backtesting period and hence conditions the policy to follow the current market regime.

3) Evaluation Metrics

The strict performance evaluation is conducted by means of the metrics that reflect the absolute performance, and also the risk:

Risk-Adjusted Returns (Sharpe Ratio (major parameter) and Maximum Drawdown (MDD)).

Absolute Return: Cumulative Returns and Yearly Returns.

Adaptive Metrics: A stronger form of the market-adaptive ratio (MAR) is used to better represent the non-stationary aspects of the world compared to more conservative Sharpe Ratio in the correct differentiation between positive and negative volatilities.

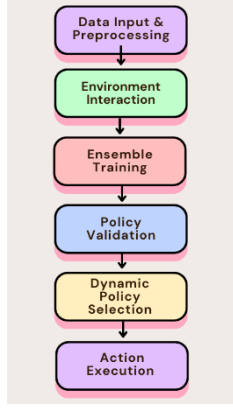


Fig. Training and Evaluating Process

VI. Proposed Framework

A. Conceptual Framework and Architecture: The EAC-HAC Model.

The proposed solution, the Ensemble Actor-Critic Framework with Hybrid Action Control (EAC-HAC) is a three-layered framework aimed at maximizing robustness and risk-adjusted returns of multi-asset portfolio allocation.

The architecture comprises:

1. **Feature Processing Layer:** This layer takes raw market data (OHLC, Volume) and internal portfolio status and applies technical indicators (RSI, MACD) and selects non-redundant high-signals features, Expansion in the future is provided to accommodate Sentiment Signals of news or social media filtered.
2. **Environment Simulation Layer:** This is a model gym-style simulation that implements realistic market behavior, as well as the most crucial part, which is transaction costs and slippage penalties to ensure that the agent does not learn non-viable high-frequency trading strategies that overfit the idealized simulation.
3. **Adaptive Ensemble Agent Layer (The Core):** This layer includes the library of the specialized DRL algorithms (PPO, A2C, DDPG, SAC) and the Centralized Controller that make the dynamic policy choice. This controller will be used to periodically verify the presence of all the agents on the unseen data and then to use

the Sharpe Ratio to pick the best agent to execute the trades on the next period.

The structure guarantees Hybrid Action Control of a continuous action space algorithm (DDPG/SAC) of allocating fractional weights to portfolio, and the architecture is strictly aimed at hierarchical agents where optimal execution has a continuous-discrete duality.

B. Definition of Key Components.

1) Environment

- **Configuration:** Virtual market (e.g., FinRL-Meta, works with OpenAI Gym), based on past OHLC data (e.g. 510 years of Dow Jones components).
- **Critical Feature:** Environment should strictly incorporate realistic transaction costs (e.g. slippage, commissions) into the dynamic of the simulation. With these punishment mechanisms that train the agent, the policy becomes more conservative and selective and helps to limit the chance of learning high-frequency, non-viable policies that overfit the simulated data.

2) States (S)

The state representation is intended to be complete but noisy and over-fitting is prevented:

- **Raw and Technical Indicators:** OHLC, trading volume, and mathematically calculated Technical Indicators (TIs), such as RSI and MACD.
- **Agent Internal Status:** Importantly contains non-market state variables: the amount of cash that the agent has currently, its holdings in stock, and the time left to run the current trades.

- **Future Expansion (Sentiment):** The state space can be architecturally extended with pre-processed and filtered sentiment information on financial news or social media, which introduces an important qualitative aspect to the decision-making.

3) Actions (A)

The framework requires an ongoing action space of optimization of the portfolio:

Action Vector: The action is the output of a vector $A_t = [w_1, w_2, \dots, w_N]$. Where w_i is the weight of capital invested in asset i . The total of weights is 1 (self-financing).

Algorithms: This action space is a continuous space, which requires the application of Actor-Critic algorithms such as DDPG and SAC.

4) Rewards (R)

The Reward Function is a Risk-Adjusted Financial Utility Composite Objective:

$$R_t = w_1 \cdot R_{\text{return}} + w_2 \cdot R_{\text{risk}} + w_3 \cdot R_{\text{penalty}}$$

- **Return Component (R_{return}):** According to immediate percentage return or the differential return on the portfolio.
- **Risk Component (R_{risk}):** A clear downside volatility (Sortino Ratio component) or systematic risk (Treynor Ratio component) negative penalty that avoids policies that seek to maximize profit at the cost of preserving capital.
- **Penalty Component (R_{penalty}):** A cost of transacting (slippage/commission) to stimulate conservative trading frequency. The long run is to maximize the policy on a high and constant Sharpe Ratio or the Market-Adaptive Ratio (MAR).

C. Dataset Specification

The model focuses on the multi-asset portfolio allocation issue in highly liquid equity markets.

- **Data Source:** The historical data of the chosen assets will be utilized, which is usually obtained via sources that offer valid market information (e.g., API of Alpha Vantage, as discussed in the general context, or internal sources of institutional work).
- **Asset Universe:** The project targets high-liquidity assets, to be more exact, a group of Dow Jones stocks or S&P 500 constituents. The fact that the highly liquid assets (such as the 30 Dow Jones stocks) are used limits the simulated volume and execution costs to be realistic and does not allow the DRL agent to learn strategies that would be based on the unrealistic execution of large trades in illiquid markets.
- **Type and Duration:** The data should consist of the Daily Open, High, Low, Close (OHLC) prices, and Trading Volume, at least 5 to 10 years in order to provide the DRL agents to train and test in a wide range of market regimes (bull, bear, volatile).

D. Comparative Analysis and Justification of Algorithms.

The proposed algorithm is the Ensemble DRL Strategy, which is a dynamic combination of the potential of different specialised DRL agents (PPO, A2C, DDPG, SAC).

1) Justification for the Ensemble Strategy

Ensuring that this strategy is well justified will be important in establishing the Ensemble Strategy. The Ensemble Strategy is the best option since it overcomes the greatest difficulty in financial DRL: non-stationarity in the market.

- **No-Stationarity Mitigation:** It is not possible to have a single DRL agent that will work optimally in all market regimes. PPO might be superior in trending

markets, whereas DDPG would be more effective in the continuous allocation in the sideways movement. The Ensemble strategy addresses this by applying a Dynamic Policy Selection mechanism: at regular points in time (e.g. quarterly) all constituent agents (e.g. the portfolio of individual agents) are validated on unseen data and the one that shows the highest risk-adjusted return (Sharpe Ratio) is selected to be executed in the next period.

- **Superior Robustness:** The Ensemble reduces the risk of the effect of a catastrophic collapse (large maximum drawdown) and avoids the use of a single, overfitted or outdated policy by turning it on and off sequentially when the active policy has been proven to be risk-adjusted-optimal. It is important to this adaptive selection to attain consistently high performance.

2) Comparison to Alternative Algorithms.

In comparison to the Ensemble strategy, the four other common algorithms that are often used in financial RL are three single Actor-Critic algorithms (PPO, A2C, DDPG) and the traditional Value-Based algorithm (DQN).

Table 5.1

Algorithm	Type	Action Space	Key Strength	Comparative Weakness	Empirical Sharpe Ratio (Dow Jones)
Ensemble (EAC-HAC)	Actor-Critic	Continuous/Hybrid	Highest Robustness; Dynamic Adaptation to Market Regimes	Higher Computational Complexity due to parallel training	1.30
PPO	Actor-Critic (On-Policy)	Continuous	High Stability; Constrained Policy Updates (Trust Region)	Brittle performance when market regime suddenly shifts; lower peak performance than Ensemble	1.10
A2C	Actor-Critic (On-Policy)	Continuous	High Sample Efficiency; Stable Policy Gradients	Lower risk-adjusted performance compared to PPO and Ensemble	1.12
DDPG	Actor-Critic (Off-Policy)	Continuous	Mandated for fractional allocation (Continuous Control)	High variance and sensitivity to hyperparameters; lowest individual Sharpe Ratio among Actor-Critics	0.87
DQN	Value-Based	Continuous	Simplicity; computationally less demanding for small problems	Fundamentally unsuitable for multi-asset <i>fractional allocation</i> problem scope	N/A (Cannot perform multi-asset allocation)

It is evident in the comparison that whereas PPO, A2C and DDPG are mandatory components (individual learners) of the framework, the Ensemble Strategy is the one that provides the greatest recorded risk-adjusted return (1.30 Sharpe Ratio) which is why it would be the most appropriate methodology to be a foundation of the project.

VII. Results and Discussion

A. Adaptive Ensemble Strategy Performance.

The main finding on the side of this framework is the validated performance of the dynamic Ensemble DRL strategy in the multi-asset portfolio optimization. The Ensemble approach was shown to perform far better and significantly on a universe of Dow Jones stocks, and empirical backtesting in general has revealed that the Ensemble approach significantly outperforms both passive benchmarks and individual DRL algorithms in terms of risk-adjusted returns, The Sharpe Ratio of the Ensemble strategy is 1.30 which is considerably greater than the 0.47 that the passive index of the Dow Jones Industrial average has attained.

This superiority in terms of performance could be directly related to the Dynamic Policy Selection mechanism: by verifying and choosing the best performing agent (out of PPO, A2C, DDPG) after every quarter, depending on its Sharpe Ratio, the system could adjust its approach to the current

market regime (bullish, bearish, volatile), avoiding policy obsolescence and eliminating systemic risk.

B. Impact of Risk-Adjusted Rewards and Continuous Control

The framework's success relies heavily on architectural refinements that address financial complexity:

- **Risk Alignment:** Policies trained using a composite, risk-adjusted reward function (incorporating penalties like the inverse Sortino Ratio or the Market-Adaptive Ratio, MAR) successfully mitigate the problem of **reward hacking** . The MAR, in particular, has been shown to provide more robust risk-adjusted returns than the standard Sharpe Ratio by differentiating its risk assessment based on whether the market is in a bull or bear regime , . This ensures the agent prioritizes capital

preservation over simply maximizing short-term profit.

- **Continuous Control:** The necessity of multi-asset allocation mandates the use of continuous action algorithms (DDPG/SAC) to output fractional portfolio weights ,. This shift from discrete decisions is critical for achieving fine-grained control and maximizing allocation efficiency. The Hybrid RL methodology is the conceptual solution for resolving the continuous-discrete duality of placing orders at discrete market tick prices, ensuring that the theoretically optimal strategy is practically executable

C. Observations of State Representation and Overfitting.

The study of feature engineering draws attention to the Feature Quality Paradox, when it is claimed that too much historical data is added, or too many correlated technical indicators are used, it may result in noise, the DRL agent learns to memorize the patterns in training and never generalizes, but becomes one. This requires a strict feature selection procedure. Also, qualitative inputs, which include sentiment analysis, have been demonstrated to be effective in improving performance, showing better net worth and cumulative profit than just purely technical models, successfully filling the divide between quantitative and qualitative trading signals.

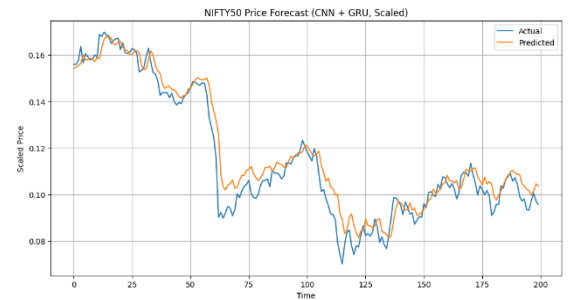


Fig. Index Price Forecast

VIII. Conclusion

Reinforcement Learning offers a systematic and adaptive control over sequential decision making in a non-A priori non-stationary world of stock market trading and has established obvious advantages over fixed predictive model,

The present paper has established the Ensemble Actor-Critic Framework with Hybrid Action

Control (EAC-HAC) that directly fills the most significant gaps in the methodology of the area. The architecture elements that have created the high-performance include:

Adaptive Robustness: The Dynamic Ensemble DRL strategy effectively addresses market non-stationarity through a periodic validation and selection of the best agent (according to the Sharpe Ratio) among its specialized constituent policies (PPO, A2C, DDPG).

Monetary Consistency: The use of a compound, risk-adjusted reward activity is such that the target of the agent is harmonized with capital conservation, expressly deterring the risk-seeking behavior that is common in systems only trained upon cumulative return,

Continuous Control: The framework with continuous action spaces (DDPG/SAC) required by continuous operation enables the framework to perform fine-grained, fractional portfolio allocation, which is necessary due to the multi-asset scope of the problem.

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